

# From point through density valuation to individual risk assessment in the discounted cash flows method.

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# Motivation

A POINT ESTIMATE CONCEALS THE UNCERTAINTY THAT MATTERS

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- The discounted-cash-flow method delivers a single number,  $V = \sum_t \mathbb{E}[CF_t] / (1 + r)^t$ , from inputs that are themselves uncertain: projected cash flows and the discount rate.
- Sensitivity tables and scenario grids, the standard practice, probe the input space pointwise and carry no probability content.
- The paper reformulates the valuation to deliver the full distribution of outcomes implied by explicit distributions on the inputs.

# Density valuation

FROM INPUT DISTRIBUTIONS TO A VALUATION LAW

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With cash-flow uncertainty  $CF_t \sim G_t$  and discount-rate uncertainty  $r \sim H$  treated explicitly, the valuation becomes a random variable

$$V = \sum_{t=1}^T \frac{CF_t}{(1+r)^t}, \quad V \sim F_V(\cdot; G_1, \dots, G_T, H),$$

- The object of interest is  $F_V$  itself: its location reproduces the classical point estimate, while its shape carries the valuation risk that the point estimate suppresses.
- The paper develops the treatment of cash-flow uncertainty and discount-rate uncertainty and their joint effect on  $F_V$ .

# | From density to individual risk assessment

## POSITION-LEVEL USE

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- Once  $F_V$  is available, position-level risk assessment is direct: quantiles of  $F_V$  bound the valuation at chosen confidence levels, and the probability of the value falling below a critical level is read off the distribution.
- The resulting risk statements integrate with individual position limits and with portfolio-level aggregation in a way point estimates cannot.
- Valuation and risk measurement collapse into one object: the same distribution answers the pricing and the risk question.

# Implications

## FOR VALUATION PRACTICE

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- Reporting a valuation as a distribution disciplines the input assumptions: every scenario grid becomes a special case of an explicit probabilistic statement.
- Disagreements between valuations can be decomposed into disagreements about input distributions, which are auditable.
- The framework extends naturally to any present-value object with uncertain flows and rates, from equity valuation to project finance.

# Thank you.

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